EXACT FUNCTORS OVER QUASI-COMPACT SCHEMES WITH AFFINE DIAGONAL ARE EXACTLY THE FM FUNCTORS WITH FLAT ACYCLIC KERNELS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original equation 0FZC: sources/stacks-chow/tags.txt, source lines 19947–19947. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GPA: sources/stacks-chow/tags.txt, source lines 20820–20820. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZG: sources/stacks-chow/tags.txt, source lines 19951–19951. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let R be a ring. Let X and Y be schemes over R. Assume X is quasi-compact with affine diagonal. There is an equivalence of categories between*

1. *the category of R-linear exact functors $F : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ which commute with arbitrary direct sums, and*
2. *the full subcategory of $QCoh(\mathcal{O}_{X \times_R Y})$ consisting of $\mathcal{K}$ such that*
   (a) *$\mathcal{K}$ is flat over X,*
   (b) *for $\mathcal{F} \in QCoh(\mathcal{O}_X)$ we have $R^q pr_{2,*}(pr_1^*\mathcal{F} \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K}) = 0$ for $q > 0$.*

given by sending $\mathcal{K}$ to the functor F in (0FZC).

Proof. Let $\mathcal{K}$ be as in (2). The functor F in (0FZC) commutes with direct sums. Since by (1) (a) the modules $\mathcal{K}$ is X-flat, we see that given a short exact sequence $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ we obtain a short exact sequence

$$0 \to \mathrm{pr}_1^*\mathcal{F}_1 \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K} \to \mathrm{pr}_1^*\mathcal{F}_2 \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K} \to \mathrm{pr}_1^*\mathcal{F}_3 \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K} \to 0$$

Since by (2)(b) the higher direct image $R^1\mathrm{pr}_{2,*}$ on the first term is zero, we conclude that $0 \to F(\mathcal{F}_1) \to F(\mathcal{F}_2) \to F(\mathcal{F}_3) \to 0$ is exact and we see that F is as in (1).

Let F be as in (1). Let $\mathcal{K}$ and $t : F \to F_{\mathcal{K}}$ be as in Lemma 0GPA. By Lemma 0FZG we see that $\mathcal{K}$ is flat over X. To finish the proof we have to show that t is an

Date: stacks-project:functors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

isomorphism and the statement on higher direct images. Both of these follow from the fact that the relative Čech complex

$$\bigoplus \mathrm{pr}_{i_0,2,*}(\mathrm{pr}_1^*\mathcal{F}\otimes\mathcal{K})_{i_0} \to \bigoplus \mathrm{pr}_{i_0i_1,2,*}(\mathrm{pr}_1^*\mathcal{F}\otimes\mathcal{K})_{i_0i_1} \to \bigoplus \mathrm{pr}_{i_0i_1i_2,2,*}(\mathrm{pr}_1^*\mathcal{F}\otimes\mathcal{K})_{i_0i_1i_2} \to \ldots$$

computes $R\mathrm{pr}_{2,*}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})$. Please see proof of Lemma 0GPA for notation and for the reason why this is so. In the proof of Lemma 0GPA we also found that this complex is equal to F applied to the complex

$$\bigoplus j_{i_0*}\mathcal{F}_{i_0} \to \bigoplus j_{i_0i_1*}\mathcal{F}_{i_0i_1} \to \bigoplus j_{i_0i_1i_2*}\mathcal{F}_{i_0i_1i_2} \to \ldots$$

This complex is exact except in degree zero with cohomology sheaf equal to $\mathcal{F}$. Hence since F is an exact functor we conclude $F = F_{\mathcal{K}}$ and that (2)(b) holds.

We omit the proof that the construction that sends F to $\mathcal{K}$ is functorial and a quasi-inverse to the functor sending $\mathcal{K}$ to the functor $F_{\mathcal{K}}$ determined by (0FZC). $\square$